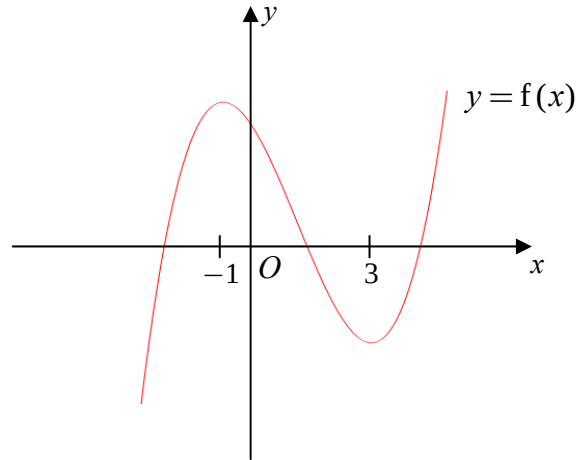


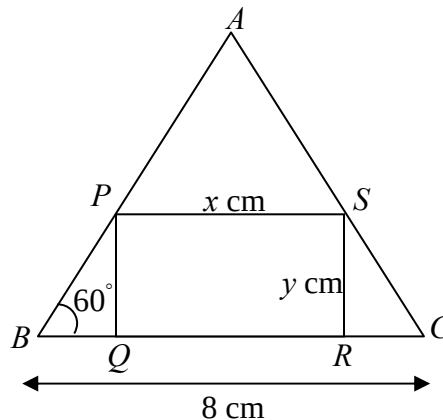
## 2009 JC 2 H1 Math Preliminary Examination

## Section A: Pure Mathematics [40 marks]

1. The diagram below shows the graph of  $y = f(x)$ . The graph has a maximum point at  $x = -1$  and a minimum point at  $x = 3$ .



- (a) State the set of values of  $x$  for which the graph is strictly increasing. [1]
- (b) Sketch the graph of  $y = f'(x)$ , indicating the  $x$ -intercept(s) clearly. [2]
- 2(i) Sketch the graph of  $y = \frac{1}{x\sqrt{x+1}} + 1$ , showing clearly the coordinates of any turning points and the equations of any asymptotes. [3]
- (ii) By adding a suitable line, solve the inequality  $\frac{1}{x\sqrt{x+1}} \geq x - 3$ . [2]
3. An equilateral triangle  $ABC$  of side 8 cm is to have a rectangle  $PQRS$  of length  $x$  cm and breadth  $y$  cm inscribed in it as shown below.



(i) Show that  $y = \frac{\sqrt{3}}{2}(8-x)$ . [2]

(ii) Hence, find the dimensions of the rectangle  $PQRS$  for which its area,  $L$  is a minimum, leaving your answer in exact form. [4]

4(a) Solve  $1 + \sqrt{e^x + 5} = e^x$ , leaving your answer in exact form. [3]

(b) Solve the simultaneous equations

$$3^{x-2} \cdot 9^y = 243$$

$$\log_4(6x - y) = 2.$$

[4]

5. The functions  $f$  and  $g$  are defined as follows:

$$f : x \mapsto \frac{1}{2x}, \quad x \in \mathbb{R}, \quad x \leq m,$$

$$g : x \mapsto \sin x, \quad x \in \mathbb{R}, \quad -\frac{\pi}{2} \leq x \leq \frac{\pi}{2},$$

where  $m$  is a negative real number.

(i) On a clearly labelled diagram, sketch the graph of  $f$  and state its range, expressing your answer in terms of  $m$ . [3]

(ii) Find the inverse function of  $f$ . [3]

(iii) State the condition under which the composite function  $g^{-1}f$  exists, and determine the maximum value of  $m$  for the condition to hold. [3]

6(i) Sketch the graph of  $y = e^{-\frac{1}{2}x+1}$ , showing the exact value of the  $y$ -intercept. [2]

(ii) Determine the  $x$ -coordinate of the point on the curve where the gradient is  $-\frac{1}{2\sqrt{e}}$ .

Hence, find the equation of the tangent at this point, leaving your answer in the

form of  $y = \frac{1}{2\sqrt{e}}(a - bx)$ , where  $a$  and  $b$  are values to be determined. [4]

(iii) Calculate the area of the region bounded by the curve  $y = e^{-\frac{1}{2}x+1}$ , the tangent line and the  $y$ -axis, leaving your answer in terms of  $e$ . [4]

### Section B: Statistics (60 Marks)

7. A quiz consists of two questions. In a class of 40 students, 32 students answered the first question correctly and 26 students answered the second question correctly. There are 22 students who answered either both questions correctly or both questions wrongly.

By considering a Venn diagram and letting  $x$  be the number of students who answered both questions correctly, find the probability that a randomly chosen student from the class answered both questions wrongly. [3]

8. A committee from the National Heritage Board would like to conduct a survey during the month of October 2009 on 500 people visiting museums in Singapore on their level of satisfaction with the exhibits and events held at the museums. The committee would also like to ensure that the sample of 500 is representative of the number of people who visited each museum, based on figures from the year 2008. Based on records obtained from the Singapore Department of Statistics, there were a total of 2.3 million visits made to the 5 museums in 2008 and the proportion of people visiting each museum is given below.

| Museum                       | % of visitors |
|------------------------------|---------------|
| Asian Civilization Museum    | 30            |
| National Museum of Singapore | 38            |
| Singapore Art Museum         | 21            |
| Singapore Philatelic Museum  | 4             |
| The Peranakan Museum         | 7             |

- (i) Explain why it will be difficult to carry out stratified sampling in this case. [2]
- (ii) State an alternative sampling method that will allow the committee to obtain a representative sample and describe the sampling method clearly. [3]
- (iii) Describe the meaning of a random sample and state whether the method stated in (ii) results in a random sample being collected? [2]

- 9 Three runners, James, Remus and Peter are to compete in a race. James and Remus have the same probability of winning and both are twice as likely to win as Peter. Assuming that there can only be one winner in a race,

- (i) show that the probability of James winning the race is 0.4. [1]
- (ii) find the probability that James does not win, given that Peter does not win either. [2]

The three runners decided to race with one another for three times. Assuming that the probability of each person winning remains the same for each race,

- (iii) calculate the probability that all the three races are won by the same person. [2]

The three runners are given a chance to compete again in another race. James' friend predicted that the probability of the runners winning the race depends heavily on the results of their previous race and he claims that the probability of winning will be doubled if the runner wins the previous race, and halved otherwise.

- (iv) Comment on the validity of the friend's prediction, justifying your answer clearly. [2]

10. Over a long period of time, Mr. Sim observed that the time,  $x$  minutes taken for him to walk from home to office follows a normal distribution with mean 20 minutes and standard deviation  $\sigma$  minutes.

- (i) Find the value of  $\sigma$  if he takes less than 15 minutes to arrive at his office 30% of the time. [4]

In a week, Mr. Sim works from Monday to Friday. He leaves his home at exactly 0800hrs each morning and has to reach his office by 0830hrs. By taking  $\sigma = 10$  min, find

- (ii) the probability that the total time he spends walking to office on Monday and Tuesday is more than 50 minutes. [2]
- (iii) the probability that Wednesday is the first day of a particular week where he will be late for work. [2]

- 11.** A year ago, a group of students conducted a survey on a large proportion of the school population to determine the number of hours a student sleeps each night. The students claim that their survey result showed that each student sleeps an average of 6.5 hours each night.

One year later, another group of students feel that this value is overstated and conducted another survey by randomly choosing a group of 60 students from the school population. The results of the survey is summarized below, where  $x$  denotes the number of hours each student sleeps every night.

$$\sum x = 361.2, \quad \sum x^2 = 2528.$$

- (i) Calculate the unbiased estimates of the population mean and variance of the number of hours a student in the school sleeps each night. **[2]**
  - (ii) Test, at the 5% level of significance whether the mean number of hours a student sleeps each night has been overestimated. **[4]**
  - (iii) Another sample of 60 students was chosen to test the same hypothesis in (ii), also at the 5% level of significance. Given that the population variance is 5.62, calculate the range of values of the sample mean that will lead to a rejection of the null hypothesis. **[4]**
- 12.** A company manufactures digital cameras in batches of 1000. From past experience, 8% of the cameras produced are defective. For the purpose of quality control, the manufacturer operates a sampling scheme whereby a random sample of 15 cameras is to be taken from each batch for testing.
- (i) Find the probability that there are no defective cameras in a randomly chosen sample of 15 cameras. **[2]**
- The batch will be released if the number of defective cameras in the sample is less than 3 and will be rejected if there are more than 3 defective cameras. If there are exactly 3 defective cameras in the sample, another random sample of 15 cameras will be chosen and tested. The batch will only be accepted if the second sample does not contain any defective cameras. Otherwise, the whole batch of 1000 cameras is rejected and re-processed.

- (ii) Find the probability that a randomly selected batch will be accepted. [3]
- (iii) Find the probability that 15 cameras will be tested in the sampling process of a batch of cameras. [2]
- (iv) Using a suitable approximation, find the probability that there are at least 100 defective cameras in a randomly selected batch of 1000 cameras. [5]

13. Reading is regarded as a leisure activity and Joseph wants to determine the relationship between the amount of leisure time a student, who has a hobby in reading and the amount of time he/she spends on reading in a week. He surveyed 8 of his friends who love reading and the data collected is shown below.

|                                                                |     |     |     |    |     |     |     |   |
|----------------------------------------------------------------|-----|-----|-----|----|-----|-----|-----|---|
| <b>Amount of leisure time per week, <math>x</math> (hours)</b> | 5   | 8   | 7   | 11 | 9   | 15  | 12  | 9 |
| <b>Amount of time spent reading, <math>y</math> (hours)</b>    | 2.5 | 6.5 | 4.5 | 7  | 8.0 | 9.5 | 9.5 | 7 |

- (i) Give a sketch of the scatter diagram for the data. [2]
- (ii) Calculate the value of the product moment correlation coefficient and comment on the relation between the variables. [3]
- (iii) Calculate the least squares regression line of  $y$  on  $x$  and estimate the number of hours a student who loves reading will spend on books when he/she has:
  - a. 10 hours of leisure time
  - b. No leisure time..
 Comment on the reliability of your answers. [5]
- (iv) Joseph added a ninth pair of data and calculated that the regression line of  $y$  on  $x$  is  $y = 0.347 + 0.696x$ . Given that the ninth pair of values has a value of  $x = 10$ , calculate the  $y$  value, leaving your answer correct to 1 decimal place. [3]

\*\*\*\* End of Paper \*\*\*\*